FLOWMLLAB / WEEK 11 / 01

From flow fields to defensible labels

Learning outcomes: distinguish rotation from shear and compression; explain why shock and vortex labels may overlap; fit and judge a small learned detector without pixel leakage; separate weak-reference agreement from physical validation.

Suggested 90-minute class: 15 minutes of diagnostic controls, 20 minutes of physical framing, 30 minutes in the notebook, 15 minutes of evidence critique, and 10 minutes for the exit assessment. Prerequisites are velocity gradients, supervised learning, cylinder wakes and compressible-flow branches from Weeks 1, 2, 7 and 8.

Research basis: Ehsan Roohi, Physics-audited joint neural segmentation of shocks and vortex cores: cross-solver transfer and controlled airfoil--cylinder studies (author-supplied research manuscript, 2026). The source describes joint detection in airfoil and cylinder fields; it is a manuscript under review, not a published article.

The executable lab starts with an original CPU teaching analog. Its final section reads real airfoil and cylinder research fields through retained figures and a provenance ledger: six fresh forward passes with the frozen task-preserving Harmonized Joint (HJ) shock-repair checkpoint. No new CFD or research-model training is claimed. The analytic warm-up and research evidence remain explicitly separate.

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Definitions used in this lecture

Segmentation: assigning every grid cell (pixel) a label, here shock, vortex core, or neither. A learned segmenter outputs, for each pixel, a number between 0 and 1 (through a sigmoid) or a set of numbers summing to 1 over classes (through a softmax); a threshold turns them into a mask. Binary cross-entropy (BCE) is the loss used for sigmoid outputs: for a logit z and label y in {0,1}, BCE = softplus(z) - y*z, whose gradient is sigmoid(z) - y.

Dice and IoU measure the overlap between a predicted mask P and a reference mask R: Dice = 2|P and R| / (|P| + |R|), IoU = |P and R| / |P or R|. Worked example on a 4-by-4 grid: if R has 4 cells and P has 5 cells of which 3 coincide with R, then |P and R| = 3, Dice = 6/9 = 0.667 and IoU = 3/6 = 0.5. Both are 1 for a perfect mask and 0 for no overlap; Dice is always at least IoU. Neither measures the position error of a front: a shock predicted one cell to the side of a one-cell-wide reference scores zero on both.

U-Net: a convolutional encoder-decoder. The encoder halves the spatial resolution several times while increasing the number of feature channels, the decoder doubles it back, and skip connections copy encoder features to the decoder at matching resolution so that fine detail survives. The research network of this week (Harmonized Joint, HJ) is a multi-branch encoder-decoder built on the same idea, not a standard U-Net.

Swirling strength: lambda_ci, the imaginary part of the complex eigenvalue of the velocity-gradient tensor where one exists; in two dimensions lambda_ci = sqrt(max(Q_d, 0)) with Q_d defined in the next section. Schlieren: an image of the density-gradient magnitude, used here only as a visual background.

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Vorticity is necessary context, not a core label

Let A be the in-plane velocity-gradient tensor with rows (u_x,u_y) and (v_x,v_y). Its symmetric part describes strain and its antisymmetric part describes local rotation. Planar vorticity is omega_z = v_x - u_y; dilatation is theta = u_x + v_y. A large derivative can also result from wall shear, a grid artifact or amplified sampling noise.

For the trace-free in-plane tensor, Q_d = -(u_x-v_y)^2/4 - u_y*v_x. The imaginary part of its complex eigenvalues gives swirling strength: lambda_ci = sqrt(max(Q_d,0)). In two dimensions these are algebraically linked, not two independent pieces of evidence. This is not the full three-dimensional vortex-classification problem.

Work three controls by hand: solid rotation u=-y, v=x gives omega=2 and lambda_ci=1; simple shear u=y, v=0 gives omega=-1 but lambda_ci=0; isotropic compression u=-x, v=-y gives theta=-2 and no swirling strength. The notebook verifies these identities numerically.

These gradients are unchanged by constant velocity translations on a fixed grid. That does not make the detector objective under arbitrary time-dependent rotating frames.

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A shock is more than a bright gradient

Density or pressure gradients indicate edges, not uniquely shocks. Contact layers, solid boundaries, expansion regions and numerical ringing can also produce strong signals. Compression, pressure rise and front geometry are complementary evidence. A velocity-gradient sensor alone cannot verify a gas-dynamic jump.

Across a locally planar stationary shock, audit the normal mass, momentum and total-energy fluxes: rho*u_n; rho*u_n^2+p; and rho*u_n*(h+|u|^2/2). A moving front requires velocity relative to that front. Use upstream and downstream states outside the numerically thick transition; oblique fronts require a local normal and consistent orientation.

The teaching field uses a tanh compression layer superposed with rotation and shear. Its labels are known construction regions. The superposition is not a Navier-Stokes or Euler solution and is not required to satisfy these jump conditions. Call it a shock-like diagnostic control, not a verified physical shock.

Discussion: why would a sensor appear accurate if its own thresholded gradient map were also used as ground truth? Design a separate jump audit and a solid-wall negative control before proposing an improved classifier.

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Overlapping structures need independent decisions

A vortex can intersect a shock. An exclusive softmax over background, shock and vortex prohibits this overlap by construction. Independent sigmoid foreground outputs allow two labels at one location; a background mask can then be derived from their complement. Ambiguous regions and exact solid geometry require explicit treatment rather than silent relabelling.

The principal research network is **Harmonized Joint (HJ)**, not a standard U-Net. It is a custom shared-encoder, multi-branch encoder-decoder. Specialist paths produce independent, potentially overlapping maps for shock, vortex core, wake/shear layer and expansion. Independent sigmoid decisions are essential because a physical location need not belong to only one structure. The full research workflow also keeps learned-only, physics-only and hybrid products distinct.

The task-preserving HJ variant is a constrained adaptation of this architecture. During shock repair it freezes the shared encoder and the complete vortex pathway and updates only the shock-specific path. If the frozen computation is deterministic, the vortex map before and after adaptation must therefore be identical up to machine precision. Freezing the vortex decoder alone would not provide that guarantee if its input encoder changed. By contrast, **HJ-joint** updates shared parameters and all task paths together: it may improve shock scores slightly, but it can degrade the vortex task. This preservation-versus-plasticity comparison is part of the scientific claim and must be tested directly, not inferred from a model name.

A capacity-matched U-Net is only a comparison baseline. It is not the principal HJ architecture, and the separate reconstruction exercise later in this lecture asks a different question again.

For conflicting shared gradients g_s and g_v, projection methods can remove a component when their dot product is negative. Such methods can help one task and hurt another; retain both signed changes. Physical diagnostic channels do not by themselves constitute a PDE-residual loss or a PINN.

Our local MLP has eight primitive/derivative features and two independent output decisions. It teaches overlap and evaluation, but has no spatial encoder, specialist convolutional branches or task-preserving research adaptation mechanism.

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Freeze the experiment before opening the test

The notebook generates eight training cases, two validation cases and two held-out cases from different construction seeds. Entire fields define groups. Subsampling training pixels is allowed; randomly splitting adjacent pixels from one field between training and testing is not a test of case generalization. Research trajectories, grids and solver families need analogous grouping and duplicate-input checks.

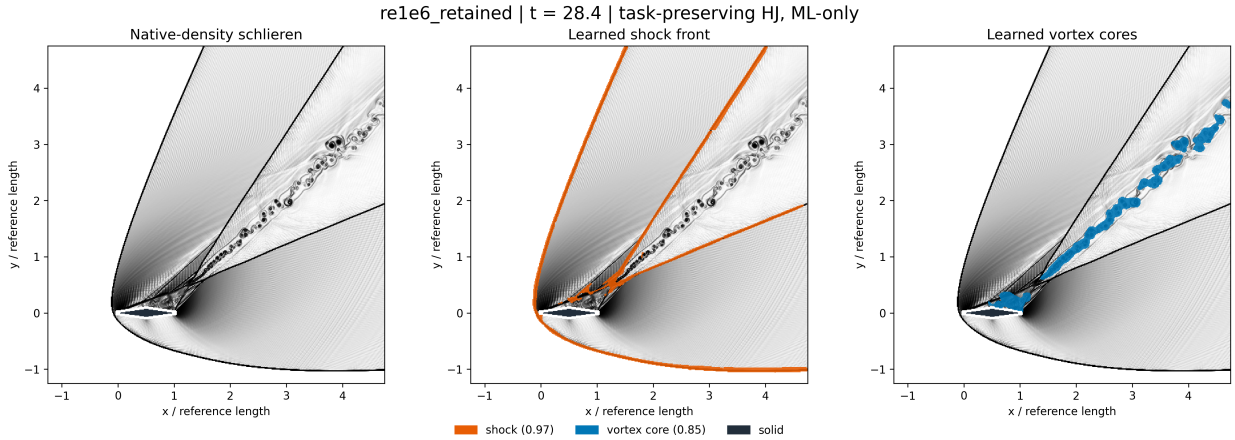
Fit the feature scaler on training pixels only. The 32-by-24 tanh MLP uses a fixed optimizer budget; it is compared with compression and swirling-strength thresholds. Select each method's two operating thresholds on validation mean case-wise Dice, then freeze them. No test-dependent quantiles, loss-weight changes or mask cleanup are permitted.

Report convergence warnings rather than hiding them. A low training loss is not physical fidelity. A physical baseline can win; that is a valid result. The notebook's two test cases provide a classroom comparison, not a statistical generalization claim across flow families.

Assignment: remove rotational inputs, rebuild the model with a fresh predeclared split, and report both task metrics and changes in fragmentation. Never reuse the already-inspected classroom test as a new blind benchmark.

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Read the colored outputs critically



Real airfoil field from Roohi's ShockVortexML study: native-density schlieren, learned shock mask and learned vortex mask. The fresh forward pass uses the frozen task-preserving HJ checkpoint `shock_repair_v2_native86_v1` and frozen thresholds 0.97 and 0.85. Only the shock and vortex outputs are displayed here; omission of the wake/shear and expansion overlays is not absence of those research paths. Geometry-crossing display-gradient stencils are excluded. These ML-only masks are not human labels or a hybrid. Thin missed branches and merged cores remain visible.

Dice = 2*|prediction intersect reference|/(|prediction|+|reference|). Empty-versus-empty is assigned one in this lab; state the convention. A front shifted by one pixel can have poor Dice while remaining close physically. Conversely, a thick predicted envelope can gain overlap without locating the centerline accurately.

The gallery retains first/middle/last eligible times for both airfoil and cylinder, excluding all human-review frames. These are previously inspected development-test trajectories; do not infer independent accuracy from appearance or mask counts. The notebook retains the synthetic Dice exercise separately. All six full-size comparisons and checkpoint provenance are in results/week11_research/.

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Reference provenance and transfer gates

The manuscript uses physics-derived proposals as weak reference masks, not an independent exact oracle. If a hybrid uses those same proposals as support, its score measures reference-conditioned agreement. Learned-only and hybrid results must therefore remain separate. Development-inspected test fields are not untouched external tests.

Before transferring to research CFD, identify the source solver and case, native coordinates, solid mask, derivative stencil, nondimensional scales, reference construction, training-family membership and checkpoint. Preserve full-field arrays as well as visualizations. Walls and missing values must not become artificial vortices through interpolation or differentiation across the solid.

Add new controls for grid changes, noise and solver changes. Recompute derivatives from perturbed primitive velocities; perturbing only the already-derived diagnostic channels tests a different problem. A shock-overlap region must not automatically veto a real core. Thresholds expressed in raster pixels change physical size with resolution.

The research repository is https://github.com/Ehsan-Roohi/ShockVortexML. Two author-channel videos use the same fixed task-preserving model family and retained cases: [airfoil Supplementary Movie S2](https://www.youtube.com/watch?v=j9rO5j3sudA) and [cylinder Supplementary Movie S8](https://www.youtube.com/watch?v=hh3K40KRBUQ). Their descriptions identify the preserved vortex/wake/expansion outputs and link back to the research release. Videos that add the separate PM-v4 expansion branch are deliberately not cited as same-network evidence. This lecture paraphrases the author-supplied manuscript and teaches its audit principles without copying a training pipeline or redistributing unpublished CFD archives.

FLOWMLLAB / WEEK 11 / 09

Reconstruct the field before identifying structures

The real-field companion asks a different question from both the manufactured classification exercise and the HJ research detector. Suppose a sensor or storage pipeline supplies a spatially coarsened velocity field. Can a learned reconstruction recover the velocity gradients needed to identify a vortex? Our independent adaptation of a super-resolution exercise from Ricardo Vinuesa's KTH course on deep learning in fluid mechanics compares three paths: interpolation followed by a physical diagnostic; U-Net velocity reconstruction followed by that same diagnostic; and direct U-Net mask prediction. U-Net is shared only between these two reconstruction/segmentation comparisons; it is not the principal HJ detector described above, and the exercise is not a reproduction of that course's code or results.

A U-Net contracts the spatial representation to learn broader context, then expands it while concatenating encoder features through skip connections. The original architecture is described by Ronneberger, Fischer and Brox (2015), https://lmb.informatik.uni-freiburg.de/people/ronneber/u-net/. The course archive illustrates single-component turbulent-field reconstruction. Here we independently implement a smaller two-level network with two velocity inputs and either two reconstructed velocity outputs or one mask logit. No archive source code, weights or figures are redistributed, and this is not an exact

reproduction of its four-level network.

For reconstruction, minimize the mean squared difference between predicted and reference velocities after per-component training-only normalization. For direct segmentation, minimize binary cross entropy against the velocity-derived mask. These losses answer different questions. A velocity error with wavelength h produces a derivative error proportional to its amplitude divided by h; low velocity MSE therefore need not imply accurate swirling strength. Thresholding a diagnostic adds another discontinuity: a small change near the cutoff can switch many pixels.

The real inputs are previously generated FlowMLLab LBM cylinder wakes, not newly simulated data and not the hypersonic cylinder article. The 32-by-78 downstream ROI is area-reduced to 8-by-19 and bilinearly restored before either neural path sees it. Both components are available. The physical coordinates retain their true aspect ratio. The native ROI is coarse and subsampled; it is not grid-independent DNS, and its boundary is not a physical wall.

A matched protocol and a limited reference

Entire Re90 and Re110 cases provide training data; Re100 provides validation; previously inspected Re105 is the retained test. Every normalization statistic and the physical diagnostic threshold come from training only. The threshold is the 90th percentile of interior training swirling strength. Compute the reference and reconstructed diagnostics using the same ROI derivative operator, not the separately archived pre-subsampling vorticity. Omit a two-cell border from the metrics. Pure rotation and simple-shear controls verify this diagnostic independently of training.

Both networks use Adam at 0.001, 60 epochs, batch size 32 and three fixed seeds. Select each checkpoint by minimum validation loss, and the direct mask probability threshold by validation Dice on a predeclared grid. Report all seeds, not the best retained-test seed. The first seed and midpoint frame are fixed for illustration. Parameter counts differ slightly because the output heads differ; this is matched input and optimization budget, not exact parameter equality.

Read the companion notebook's table in two parts. Velocity and vorticity relative L2 errors assess reconstructed fields; divergence RMS is an additional consistency diagnostic, not an enforced constraint. Frame-macro Dice and IoU assess agreement with the thresholded native swirling-strength reference. Direct segmentation cannot have a velocity error because it produces no velocity. Empty-versus-empty masks score one. Temporal frames are correlated, so seed dispersion is not a confidence interval over independent flows.

These wakes contain vortices but no shocks. The comparison cannot establish shock detection or identify every dynamically important coherent structure. Agreement with a derivative-based weak reference is not human-validated physical accuracy. Additional higher-resolution, independent flows

and threshold-sensitivity experiments are required before broader research claims. The companion protocol records the source archive hash and explains the adaptation; the archive folder name alone does not establish code authorship or redistribution permission.

Run notebooks/week11/W11_Lab2_Reconstruction_and_Identification.ipynb to audit the retained comparison and view loss histories. Its optional retraining cell runs all six fits into a fresh scratch directory, never into retained results. The synthetic warm-up and the separate research-image gallery remain available and retain their original evidence limitations.

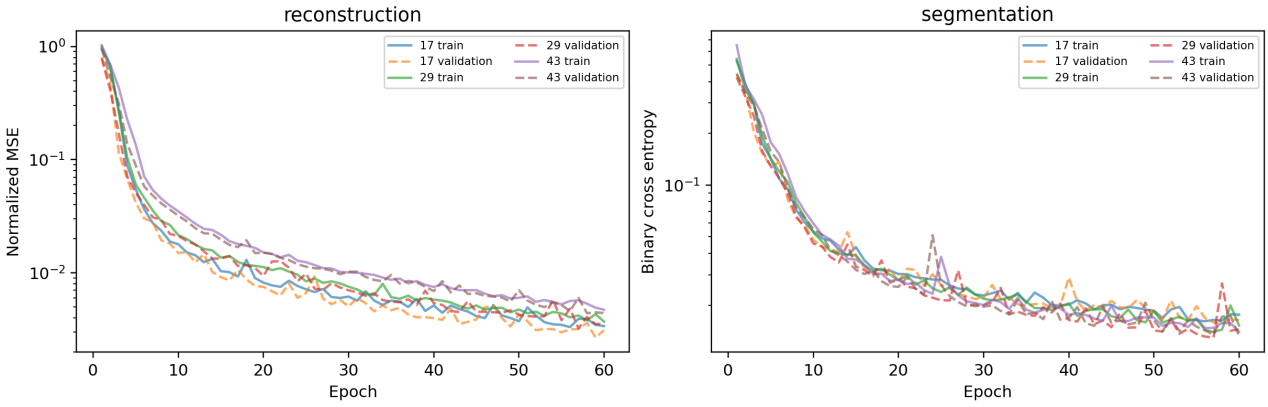
Lab 2 execution modes: the default is retained-evidence audit with no new optimizer steps. Optional small training uses seed 11, 30 epochs, learning rate 0.003, batch 8 and every twelfth frame. The retained full protocol uses seeds 17/29/43, 60 epochs, learning rate 0.001 and batch 32; the small run does not reproduce that table.

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Read the retained reconstruction comparison

Method	Velocity L2 (%)	Vorticity L2 (%)	Dice
Interpolation	7.886	39.801	0.773
U-Net field + diagnostic	1.917	12.633	0.943
Direct U-Net mask	N/A	N/A	0.971

The neural rows average three independently initialized fits on the same retained case; they are not confidence intervals across flows. The full per-seed table and color-field comparisons are in the companion notebook and results/week11_reconstruction/. All seven metric rows were recomputed from the saved models. The direct mask model scores higher against this particular weak reference, while the reconstruction model provides velocities for downstream diagnostics. Neither observation demonstrates shock detection or cross-solver generalization.



Solid curves show training loss and dashed curves validation loss. The two panels have different objectives and cannot be ranked by comparing their absolute heights. Several best checkpoints occur near the end of the fixed budget: this is a completed controlled run, not a proof of optimization convergence. Extending the budget is a new predeclared experiment, not an opportunity to select a better retained-test result retrospectively.

Exit assessment and reproducibility

Submit one page with: the split IDs, frozen thresholds, both methods' case-wise Dice, at least one failure, and a labeled comparison figure. Explain why vorticity does not imply a vortex, why independent sigmoid labels are appropriate, and why physical-label agreement is not independent physical validation.

Propose a front-location metric and a component-matching rule before looking at a new research case. State how the rules handle geometry, ambiguity, missing structures and empty references. Describe the stronger evidence needed for cross-solver claims and for temporal vortex tracking.

Run notebooks/week11/W11_Shock_Vortex_Identification.ipynb from a complete FlowMLLab checkout or use its Colab launcher. It runs on CPU and keeps experiment outputs in memory. The authoring builder can regenerate the retained teaching notebook and figure; a student's Run All does not overwrite results/.

Source attribution: Ehsan Roohi, Physics-audited joint neural segmentation of shocks and vortex cores: cross-solver transfer and controlled airfoil--cylinder studies, author-supplied 2026 manuscript; ShockVortexML repository above. Original FlowMLLab lecture wording, synthetic fields and code were AI-assisted and checked. These additions are not claims of original research-data generation or full-paper reproduction.

Hydrofoil cavitation: the machine-vision extension

The final Week 11 example uses our existing hydrofoil vapor-cloud detection code and results. The task is to distinguish background, attached cavity and disconnected vapor cloud in a two-dimensional CFD field. A vapor-fraction image shows where vapor is present; attachment also depends on connectivity and contact with the hydrofoil. A thin ligament can change the interpretation of a large region.

Unlike the overlapping shock and vortex labels, attached and disconnected are mutually exclusive states under the stated topology definition. A three-logit softmax classifier is therefore appropriate for this experiment. Solid and uncertain support need separate handling. A disconnected region in a two-dimensional field is not evidence of three-dimensional disconnection or a tracked shedding event.

This is Ehsan Roohi's retained native_alpha20_v6 experiment from September 12, 2026, packaged as an executable extension. Inputs are real Fluent CFD vapor-fraction rasters and geometry, not experimental photographs. We reuse the original trained weights and 158 snapshots from seven cases. Default notebook execution reruns inference; it does not generate new CFD or retrain the model.

Allow 30-45 additional classroom minutes: inspect the input and reference definition, run the saved detector, compare the cavity field and its detected structures, then discuss one failure. The companion is notebooks/week11/W11_Cavitation_Cloud_Detection.ipynb; the course launchers provide its direct Colab link.

Vapor support, native topology and learned classes

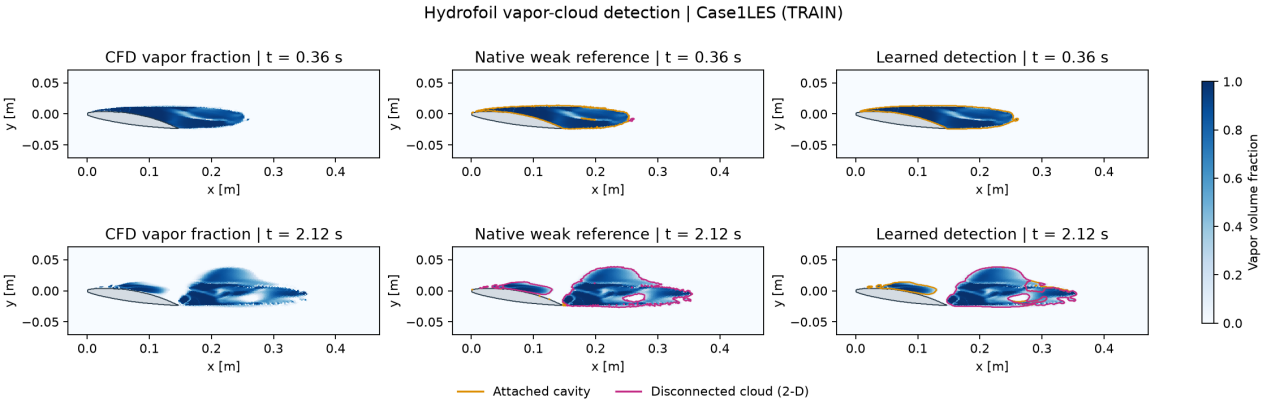
The native weak teacher defines vapor support by alpha_v >= 0.20. Connected components are computed using original Fluent cell-face adjacency. A component that touches an actual wall face is attached; a component without that contact is disconnected in the two-dimensional representation. Components affected by external or cropped support remain uncertain. The published reference arrays retain that uncertainty as label 255, together with solid support excluded from scientific scores.

The model receives three channels: vapor fraction, the solid mask, and the clipped pixel distance to solid, min(d/48,1). The five encoder widths are 8, 16, 24, 32 and 48. A learned projection of global-average bottleneck features supplies context; four skip-connected decoder stages use bilinear upsampling. A three-channel output head gives background, attached and disconnected logits. The original model has 126,275 trainable parameters.

Inference is direct neural argmax. Reference masks do not enter inference, and no connectivity repair, hand-filled gap, component deletion or geometry veto is added afterward. Wall false positives are reported separately rather than silently hidden. The pixel-distance channel preserves the original preprocessing; it is not invariant to changing raster resolution.

The references are algorithmic, not human ground truth. A separate classroom baseline thresholds the raster and uses four-connected pixels with one-cell wall contact. Its disagreement with the native teacher illustrates why raster connectivity must not be substituted silently for native-mesh contact. Neural predictions and this classical baseline remain separate products.

See the cavity and its detection together



Left: CFD vapor volume fraction. Middle: native weak reference. Right: fresh inference from the original model. Orange denotes attached cavity; magenta denotes disconnected vapor in 2-D. Every panel uses the same vapor range [0,1] and physical coordinates. The illustrated times, 0.36 and 2.12 s, come from the TRAIN LES trajectory and were retained in the original review.

The large downstream cloud is an interpretable detection, but the near-wall attachment decision can still be wrong. Inspect both classes, including small ligaments and wall errors. Do not interpret a visually persuasive training example as independent test accuracy. The complete gallery retains the original four LES times and worst-error examples.

Reproduce the result and keep the difficult cases

The case split is fixed: Cases 13, 14, 16 and Case1LES are TRAIN (86 frames); Case 19 is validation (27); Cases 24 and 23 are nontraining (27 and 18). All were previously inspected during development. Neighboring frames are correlated; 158 frames do not represent 158 independent flow cases. No fresh blind evaluation is claimed.

Rerun all frames with the saved model and compare every predicted pixel to the archived argmax output. Then recompute pooled TP, FP and FN over valid reference support. Dice = 2*TP/(2*TP+FP+FN). Empty-versus-empty is undefined here, unlike the synthetic warm-up's declared convention. Also count predictions inside solid and run a zero-vapor negative control with unchanged geometry.

For Case1LES, attached/cloud Dice is 0.93323/0.88713. For inspected nontraining Case 24 it is 0.96270/0.49188, while Case 23 gives 0.97995/0.85525. The lower Case 24 cloud score remains part of the lesson. All seven cases, confusion counts, data hashes and figure times are retained in results/week11_cavitation/ and data/week11_cavitation/manifest.json.

The notebook additionally compares a transparent raster baseline and adds seeded vapor-channel noise with standard deviation 0.02. This is a sensitivity exercise against the same clean reference, not a camera-noise model or an independent validation experiment. A baseline may win; do not tune the detector to conceal that outcome.

Training provenance, interpretation and assignment

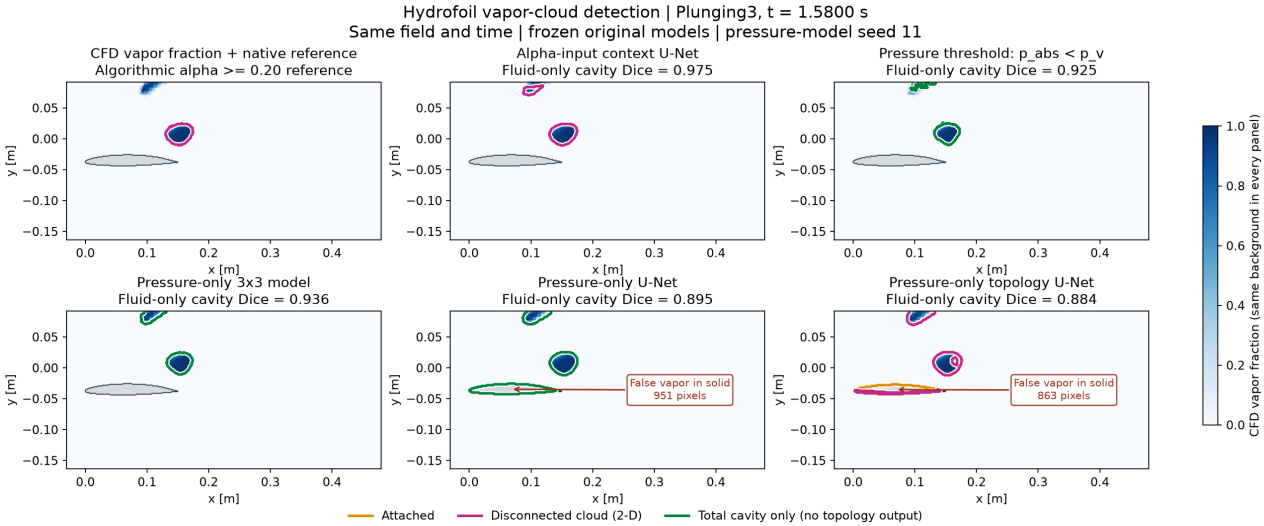
The retained v6 model starts from native_v4, whose earlier training used alpha_v >= 0.50 references, and receives 1000 additional Adam updates at learning rate 0.0002 with alpha20 labels. Its parent had 2000 updates. A poor from-scratch alpha20 development fit preceded this continuation. Report that history; do not call v6 a 1000-step from-scratch model or an equal-total-budget comparison.

The optional notebook section reruns this final adaptation from the supplied parent using only the four TRAIN cases. It preserves cloud-frame sampling, vertical reflection, class-weighted cross entropy plus foreground Dice, and gradient clipping. The fixed final step is retained; no PDE residual or forecasting objective is used. A separate in-memory model is returned, leaving the original weights unchanged. Raw Fluent conversion and the earlier parent training are outside this public subset.

Assignment: identify one attachment error in Case 24; compare neural and raster-baseline scores for both classes; explain the effect of an unresolved vapor ligament; and specify the new trajectories, expert annotations and temporal sampling needed to assess true cloud detachment and tracking. A smooth outline or a high weak-reference Dice alone does not validate those claims.

Related work: Hatzissawidis et al., Deep learning semantic segmentation for cloud cavitation image analysis, Physics of Fluids 38, 093331 (2026), doi:10.1063/5.0345365. That study segments camera images and uses a subsequent heuristic sheet/cloud split. Our retained experiment directly predicts three classes from CFD rasters. This lesson does not reproduce their dataset or establish superiority to their method. Course adaptation and verification are AI-assisted; exact author-model and data provenance accompany the code.

Alpha and pressure methods on the same field



The six panels show one retained CFD frame and five original methods: alpha-input context U-Net, fixed pressure threshold, pressure-only 3x3 model, pressure U-Net and pressure topology U-Net. All panels share geometry, physical coordinates, time and the same alpha background. Alpha is displayed for comparison but never enters pressure inference.

Orange and magenta mean attached and disconnected classes. Green means total cavity for methods without a topology output. White halos improve contour visibility. Arrows mark false vapor inside the solid hydrofoil, not attached cavities. Fluid-only Dice excludes solid and uncertain support, so those solid errors are counted separately. Common cavity Dice unites the two topology classes and compares that support with scalar predictions thresholded at 0.20.

The plot uses the first pressure seed (11) and the Plunging3 frame with maximum valid CFD cavity area, independently of model scores. The notebook replays all 16 frames from Plunging3 and Oscillation3 with all three pressure seeds. Both trajectories were excluded from optimization but previously inspected. Neither contains valid attached-reference pixels, so attached-class transfer is not validated here.

Interpreting pressure-based detection

The input deficit is gauge pressure plus operating pressure minus configured vapor pressure, in Pa. Preserve the original float32 operation order. Four asinh channels use scales 10, 100, 1000 and 10000 Pa, together with geometry. The fixed pressure baseline uses absolute pressure strictly below configured vapor pressure; it is a total-cavity comparator, not a topology classifier.

The 3x3 model uses local pressure and wall patches with two small tanh layers. The pressure U-Net uses wider spatial context and skip connections. The topology U-Net adds a three-class head. These are the original saved networks, including weaker variants. Default execution performs inference only. The pressure models have no vapor-fraction input; the alpha model has different information and training history, so this is not a matched-input architecture ranking.

More context did not uniformly improve the original results. On Plunging3, the original three-seed pressure-patch Dice fell from 0.8711 for clean pressure to 0.7357 with 1000-Pa input-noise standard deviation. Oscillation3 fell from 0.9109 to 0.8181. These historical noise results remain in the manifest; the new gallery replays clean inputs. A pressure deficit is a useful feature, not a cavitation transport law or a learned shedding forecast.

Inspect the visible wall false positives, missed cloud edges and the distinction between frame and trajectory scores. Native references are algorithmic and exclude solid/uncertain support from Dice, while the overlays retain unmodified predictions. The two examples do not establish temporal object identity, new-geometry transfer or independent physical accuracy.

Run the final sections of W11_Cavitation_Cloud_Detection.ipynb. The reusable implementation is flowmllab/cavitation_methods.py; numeric weights, source hashes and pressure settings are in data/week11_cavitation_methods/. Explain why the pressure threshold can disagree with vapor fraction, and what independent evidence would be needed before choosing a detector for a new flow.